RESEARCH BRIEF / KDD 2024

FinAgent

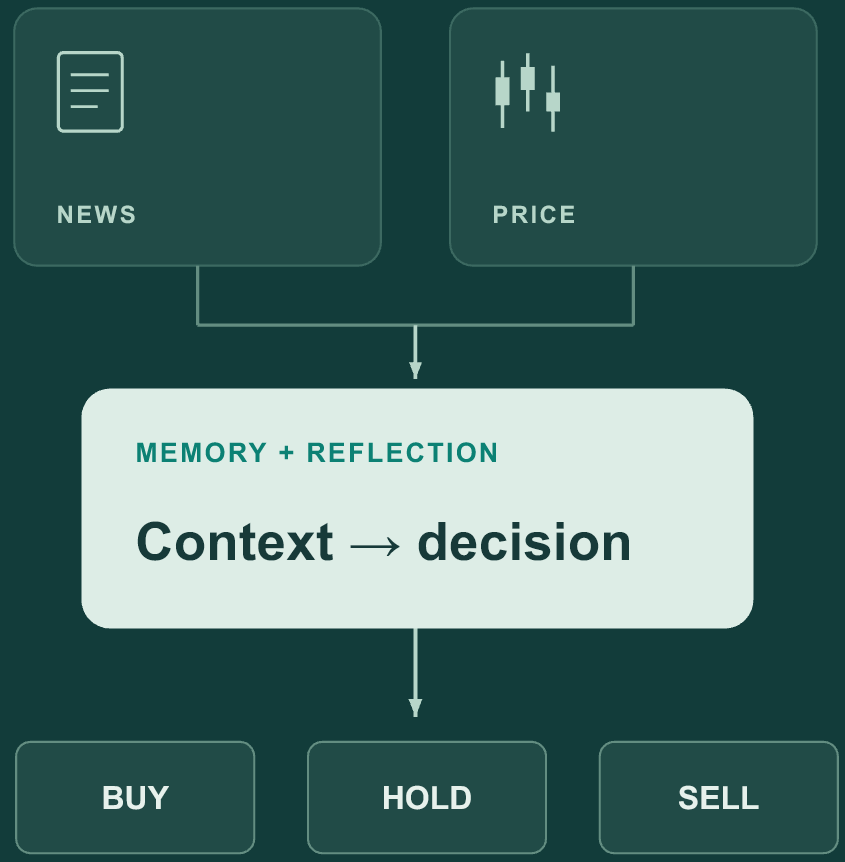
A Multimodal Foundation Agent for Financial Trading

Tool-Augmented, Diversified, and Generalist

> **Beyond forecasting: turn market evidence into action.**

Wentao Zhang et al. · KDD '24, pp. 4314–4325

MULTIMODAL IN. REASONED ACTION OUT.

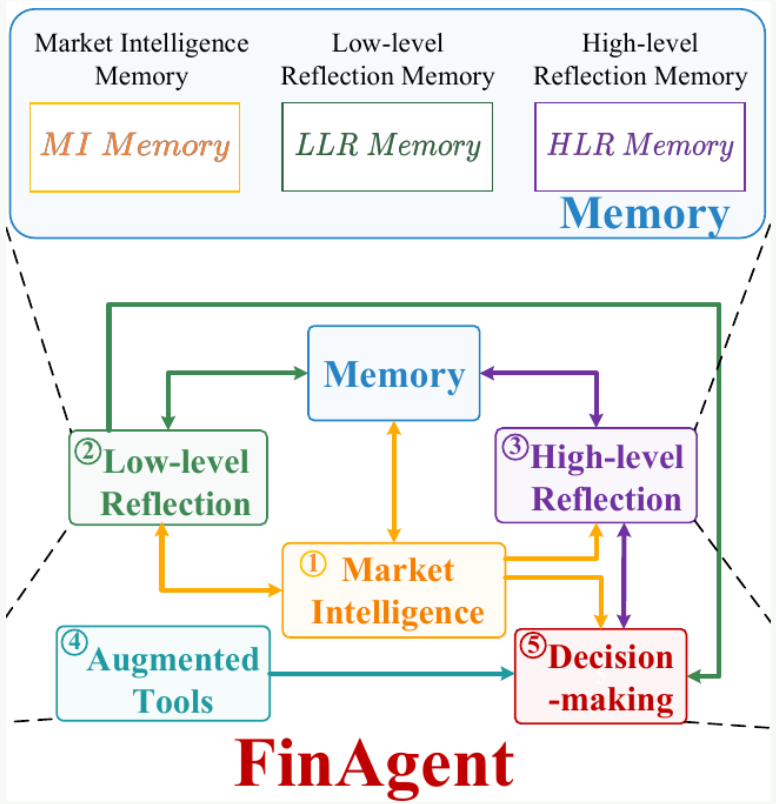


Observe. Recall. Reflect. Act.

Conceptual illustration adapted from the paper.

FinAgent decision architecture

Core workflow from the paper's original Figure 2



Market intelligence

Summarize current market information.

Memory and reflection

Retrieve relevant experience and review market signals and previous trades.

Decision with tool support

Combine evidence with technical strategies, expert guidance and trader preferences.

Source: Zhang et al., Figure 2, arXiv:2402.18485v3, p. 5. Original figure cropped to its central workflow.

Two reflections. Two different questions.

Separate market interpretation from evaluation of the agent's own behavior.

01 LOW-LEVEL / THE MARKET

Why did prices move?



Market information ↔ price movements

SHORT | MEDIUM | LONG

02 HIGH-LEVEL / THE TRADER

Was the trade sensible?



Past action + rationale + outcome

EXTRACT LESSONS

DIVERSIFIED RETRIEVAL

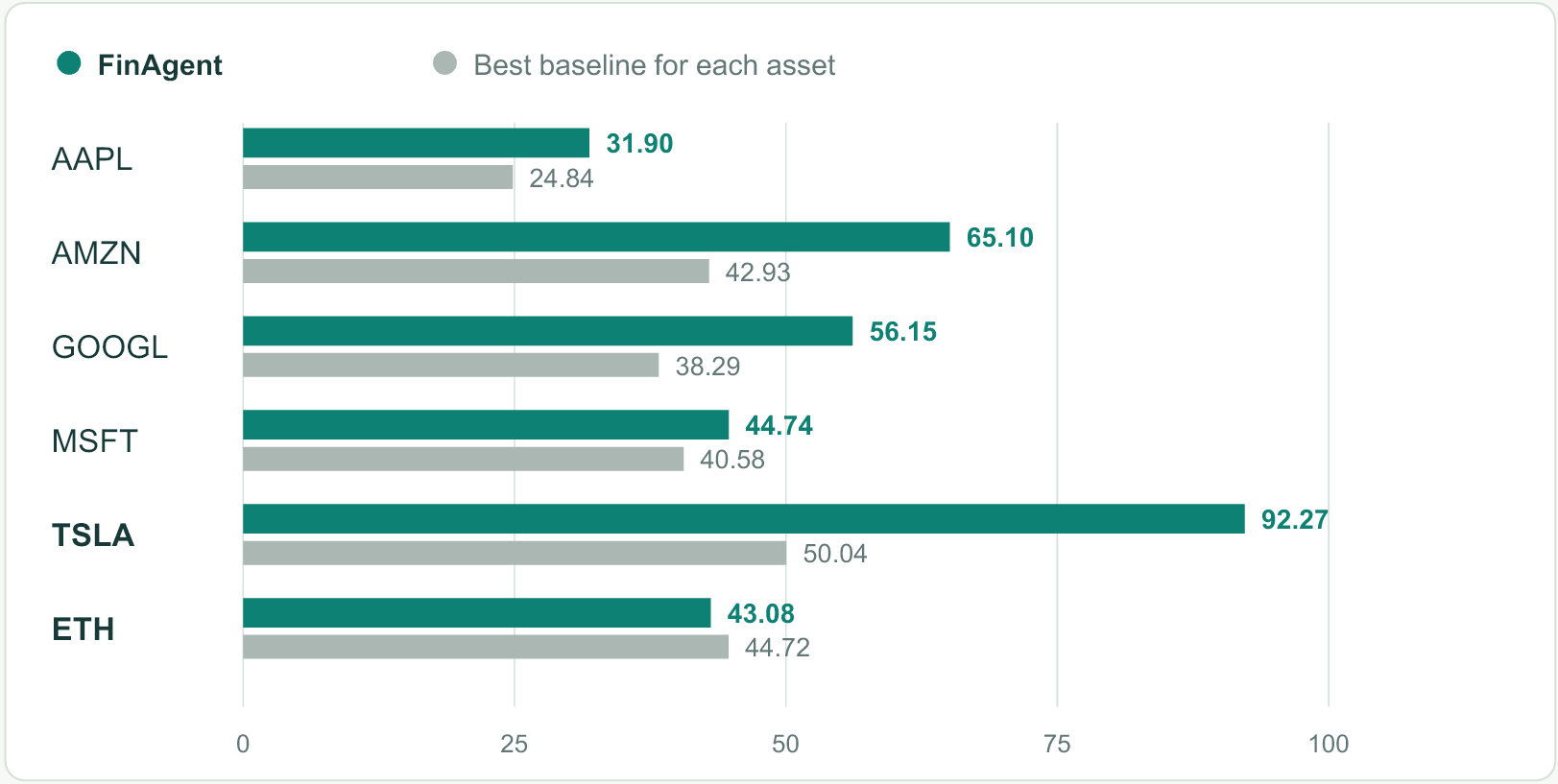
Decision summary → Trading decision

Retrieval query → Relevant history

Source: Zhang et al., KDD 2024, §4.1–4.3. Questions and diagram are explanatory paraphrases.

Strong on stocks. Not best on every asset.

Annualized return (ARR, %) · Test window: 1 Jun 2023 – 1 Jan 2024



SPOTLIGHT / TSLA

92.27%

annualized return for FinAgent

50.04%
for FinMem

ETH: FinMem is ahead.

5 stocks + Ethereum · 12 baselines · Historical backtest, not a live-trading guarantee

Source: Zhang et al., KDD 2024, Table 4 and §5. Best-baseline identities are listed in the hidden appendix.

Tool effects differ across assets

Component ablation from the original paper. Annualized return (ARR, %)

ORIGINAL TABLE 5

M	L	H	T	TSLA ARR%↑	TSLA SR↑	TSLA MDD%↓	ETHUSD ARR%↑	ETHUSD SR↑	ETHUSD MDD%↓
			√	39.01	0.90	22.54	16.21	0.63	15.93
√				39.27	0.77	30.15	25.97	0.77	24.43
√	√			57.16(+45.56%)	1.02 (+33.14%)	25.77(-14.52%)	52.33(+101.48%)	1.34(+72.99%)	13.59(-44.39%)
√	√	√		89.25(+56.14%)	1.46(+42.86%)	27.62(+7.18%)	54.80(+4.73%)	1.40(+5.09%)	11.74(-13.57%)
√	√	√	√	92.27(+3.38%)	2.01(+37.84%)	12.14(-56.04%)	43.08(-21.39%)	1.18(-16.09%)	12.72(+8.30%)

The last two rows isolate the addition of tools

M: market intelligence L: low-level reflection

H: high-level reflection T: augmented tools

The auxiliary strategies were designed for stocks. Their effect differs on Ethereum.

AFTER ADDING TOOLS

Market fit matters.

TSLA +3.02 pp

ETH −11.72 pp

pp = percentage points of ARR

Source: Zhang et al., Table 5, arXiv:2402.18485v3, p. 9. Original table crop. Percentage-point changes calculated from ARR.

The contribution is the decision architecture.

Multimodal evidence. Relevant memory. Structured self-review.

01 Combine evidence

Read prices, language and visual context together.

02 Recall what matters

Retrieve relevant experience, not just more text.

03 Review both sides

Interpret the market—and evaluate prior actions.

KEEP THE CLAIM PROPORTIONATE

Promising, not proven everywhere.

Six assets. One historical window.
Further live-market validation is needed.

Zhang et al. · KDD 2024 · DOI: 10.1145/3637528.3671801 | Critical interpretation by the presenter.

Evidence behind the visuals

Backup slide — excluded from the three-minute slide show.

TABLE 4 / ANNUALIZED RETURN (%)

Asset	FinAgent	Best baseline	Method	Gap (pp)
AAPL	**31.90**	24.84	SAC	+7.06
AMZN	**65.10**	42.93	FinGPT	+22.17
GOOGL	**56.15**	38.29	PPO	+17.86
MSFT	**44.74**	40.58	FinMem	+4.16
TSLA	**92.27**	50.04	FinMem	+42.23
ETH	**43.08**	44.72	FinMem	-1.64

ARR is annualized, not the cumulative test-period return.
Baseline selection uses the highest ARR in each asset column.

PRIMARY SOURCE

Wentao Zhang et al.
A Multimodal Foundation Agent for Financial Trading: Tool-Augmented, Diversified, and Generalist.

KDD 2024, pp. 4314–4325.
DOI: 10.1145/3637528.3671801

Architecture: Fig. 2 / §4
Comparison: Table 4
Ablation: Table 5

Open the original paper ↗